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Improved Resistive Switching Observed in Ti/Zr_3N_2/*p*-Si Capacitor via Hydrogen Passivation

DONGJOO BAE, DOOWON LEE, SUNGHO KIM, AND HEE-DONG KIM
Department of Electrical Engineering and Convergence Engineering for Intelligent Drone, Sejong University, Seoul 05006, Republic of Korea

Corresponding author: Hee-Dong Kim (khd0708@sejong.ac.kr)

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ABSTRACT Charge-trap based resistive switching (RS) has attracted attention in the resistive random-access memory (RRAM) industry due to its gradual RS behavior for multi-level and synaptic applications. In this work, in order to lower the operating current level closely related to device's degradation, we applied a hydrogen passivation to Zr_3N_2 based RRAM devices and investigated the correlation between current level and trap density, such as an interface trap density (N_{it}) at the Zr_3N_2/*p*-Si layer and nitride trap density (N_{nt}) within Zr_3N_2 films, for memory cells annealed in conventional N_2 gas as well as H_2 gas. Compared to the N_2-annealed sample, after H_2 annealing, N_{it} is lowered by the hydrogen passivation effect, which results in a reduction of both current level at high resistive state (HRS) and variation of HRS and low resistive state (LRS). As a result, in the H_2 annealed Zr_3N_2 RRAM cell, we observed a lower operation voltage/current, longer endurance, and larger read margin due to the hydrogen passivation effect.

INDEX TERMS Hydrogen passivation, rapid thermal annealing, resistive switching, self-rectifying, Zr_3N_2, nitride trap density, interface trap density.

I. INTRODUCTION

Next-generation non-volatile memories (NVMs) with fast speed, low energy consumption, high scalability, long cycle endurance, and excellent retention time are essential for the development of neuromorphic devices and intelligence drones [1]–[3]. Magnetic random-access memory (MRAM), phase change random-access memory (PRAM), and resistive random-access memory (RRAM) have been extensively studied as candidates for next-generation NVMs [4]–[6]. Among them, using transition metal oxide (TMO) and nitride (TMN), a RRAM device has received enormous attention due to its simple structure, fast switching speed of >10 ns, low power consumption of <5 V, and excellent scalability [4], [7]–[14]. To explain resistive switching (RS) in RRAM devices, several possible RS models have been proposed. One of the comprehensive mechanism models is that the resistance states of the RRAM are modulated by the rupture or formation of conductive filament consisting of oxygen/nitride vacancies or metal ions [15]–[19]. Another promising model is that the resistance states of the RRAM are changed by the charge trapping process, which allows analog resistance modulation. On the other hand, when using a conductive filament model, normally digital RS for HRS and LRS is required [20]. For neuromorphic devices, charge trap-based analog RS is suitable due to its synaptic behavior [21]–[23]. Therefore, the charge trapping model has been intensively investigated recently [20], [24]. In a previous work [25], Luis V. A. Scalvi *et al.* reported that analog RS characteristics of SnO_2/TiO_2-based RRAM is dominantly modulated by charge trapping/detrapping. In another previous work [26], Ye Zhou *et al.* demonstrated the feasibility of RRAM based on charge trapping model for neuromorphic devices. In addition, since the resistance states of charge trap-based RRAM devices are modulated by charged trap density within an insulator, the correlation between the traps of interface/nitride (N_{it}/N_{nt}) and electrical properties, such as RS behavior and reliability, is identified. In our previous work [27], we reported the feasibility of Zr_3N_2-based RRAM and the correlation between N_{it}/N_{nt} and resistance states. As a result, we successfully demonstrated that the N_{nt} mainly modulated resistance states of the Zr_3N_2-based RRAM. However, the correlation

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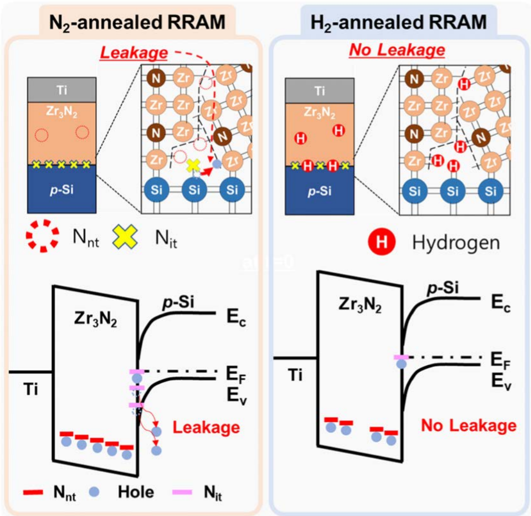

FIGURE 1. Schematic of hydrogen passivation effect on N_2- and H_2-annealed RRAM cells.

between the performance of the RRAM and N_{it} was not illustrated in detail. The generation of N_{it} in a memory device is commonly one of major failures to decrease in reliability [28]. Therefore, many groups have attempted to reduce N_{it} through post-treatment using rapid thermal annealing (RTA) in H_2 gas ambient since it has a significant effect on reducing N_{it} by passivating dangling bonds by H^+ [29]–[34]. N_{it} attributes dispersions of operation voltage and current, resulting in uniformity and retention for charge trapping model [35], [36]. However, studies for correlation among N_{it} in TMN, resistance states, and reliability have not been intensively studied in detail. In addition, numerical analysis based on precise measurement results are needed for N_{it}.

Therefore, in this paper, we investigated effects of N_{it} on RRAM devices after hydrogen passivation via the H_2 annealing process in order to reveal hydrogen passivation effect for Zr_3N_2-based RRAM, as shown in the schematic structure of Fig. 1. Especially, since holes trapped in N_{it} can release external energies such bias and heat, which results in a change in resistance state, as shown in Fig. 1, we evaluate the reliability of RRAMs. As a result, we successfully demonstrated a correlation between the N_{it} using Terman method and RS characteristic, such as operating voltage/current, endurance, retention, and read margin characteristics.

II. EXPERIMENTAL SECTION

First, the 300 μm thick p-type silicon (p-Si) wafer which was doped with 5.9 $\times$ 10^{16} cm^{-3} of Boron was cleaned with ethanol, methanol, and deionized water for 10 minutes, respectively. First, the p-type silicon (p-Si) wafer was cleaned with ethanol, methanol, and deionized water for 10 minutes, respectively. Then the Zr_3N_2 RS layer of 90 nm was deposited using reactive magnetron radio frequency (RF) sputtering at 100 W power. The sputtering chamber for Zr_3N_2 deposition was evacuated to 1.6 mTorr with Ar of 20 sccm.

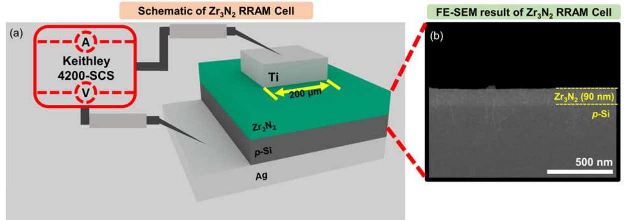

FIGURE 2. (a) Schematic illustration for the proposed Zr_3N_2 RRAM cell and measurement system, and (b) cross-section image measured by FE-SEM for the Zr_3N_2/p-Si layers.

After that, a 50 nm thick Ti with 200 μm diameter dots was patterned on the Zr_3N_2 RS layer as a top electrode. Finally, the fabricated RRAM cell was annealed at 450 ° for 30 s under 4 % of H_2 mixed Ar gas ambient with the pressure of 5 $\times$ 10^{-3} Torr to crystallize and passivate N_{it} using RTA, MILA-5000, and this is called the H_2-annealed RRAM cell. And then, Ag paste was used on the backside of p-Si wafer to be a contact metal. For reference, the additionally fabricated RRAM cell was annealed at 450 ° for 30 s in N_2 ambient, and this is called the N_2-annealed RRAM cell. A schematic drawing and cross-sectional field-emission scanning electron microscope (FE-SEM) image of the proposed Ti/Zr_3N_2/p-Si RRAM are shown in Figs. 2(a) and (b), respectively. A Keithley 4200 semiconductor characterization system (SCS)/ 4214 capacitance voltage unit (CVU) was used to measure the electrical characteristics of the RRAM cells while the p-Si was grounded.

III. RESULTS AND DISCUSSION

First of all, current-voltage ($I-V$) characteristics in dc mode were implemented to investigate the RS characteristics for N_2- and H_2-annealed RRAM cells at room temperature (RT), as shown in Fig. 3(a). In the case of the N_2-annealed RRAM cell, set and reset process was observed at +10 V and −15 V by DC bias sweeps as follows: +10 V → 0 V → −15 V → 0 V → +10 V. In this experiment, a set process in the negative bias region and a rectifying property in the positive bias region, which are typical bipolar RS characteristics of metal-insulator-silicon ZrN RRAM, were observed. This result of the N_2-annealed RRAM cell is well consistent with our previous work [27], where the set process from high resistance state (HRS) to low resistance state (LRS) is explained by hole trapping under a dc-sweep process in the negative bias region. Then, in the case of H_2-annealed RRAM cell, which also has a bipolar RS operation, the initial memory state of HRS was changed to LRS at −5 V, called set operation, while by re-sweeping a bias voltage up to +3 V, the memory state is switched back to HRS, about +0.15 V, which is called the reset operation. Meanwhile, according to the reported literature related to H_2 annealing [37], [38], it is well-known that hydrogen allows a different bonding energy depending on electronegativity of molecules such as N-H, N-O, N-Zr, and N-F, which are dipole-inducing forms.

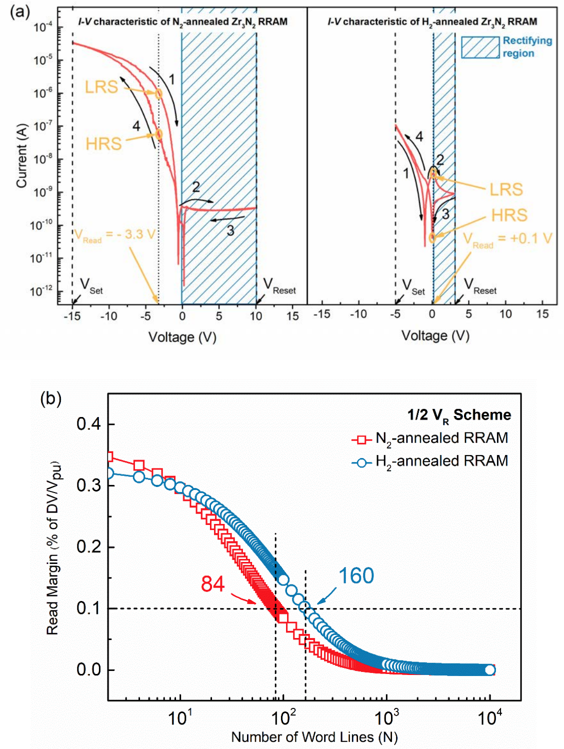

FIGURE 3. (a) Bipolar RS characteristic observed in N_2- and H_2-annealed RRAM cells, (b) Read margin for N_2- and H_2-annealed RRAM cells.

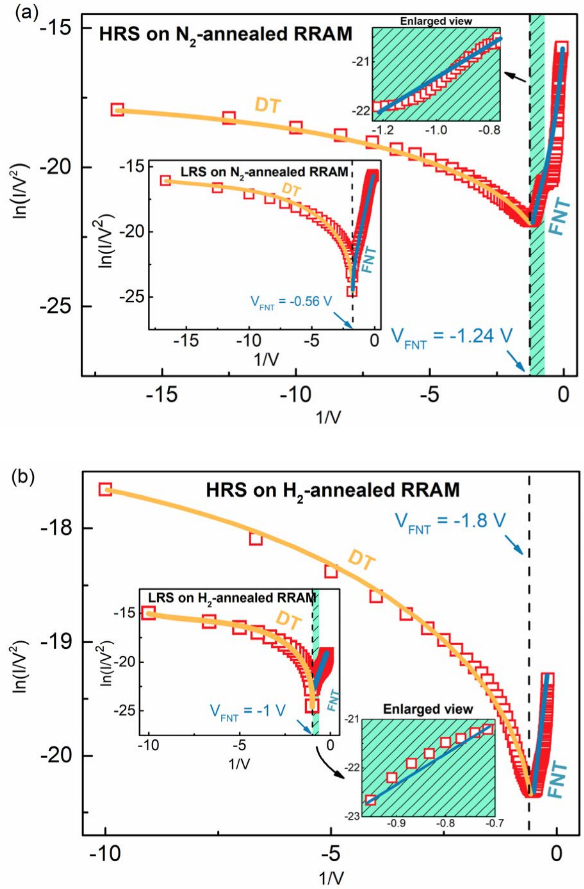

FIGURE 4. Conduction mechanism of RRAM: DT plot with I/V^2 - $1/V$ scales from I–V curve of LRS/HRS on (a) N_2- and (b) H_2-annealed RRAM cells.

Especially, in the negative bias region, the current at LRS is lower than that at HRS because of the internal electric field induced by the formed dipole, which results in a negative shift in the $I - V$ curve characteristics [39], [40]. Then, if we use $V_{read} = +0.1$ V, currents at HRS and LRS are 0.125 nA and 3.696 nA, and the current ratio (CR) between HRS and HRS is about 29.6. As a result, the H_2-annealed RRAM cell showed a lower current and voltage level than the N_2-annealed RRAM cell after hydrogen passivation, as shown in Fig. 3(a). In addition, both N_2- and H_2-annealed RRAM cells showed the current suppression phenomenon due to the intrinsic self-rectifying characteristics of *p*-Si, as shown in Fig. 3(a), which results in improvement for memory integration because the rectifying characteristic suppressed the interference [41]. Note that suppressing sneak current paths in the array is one of the crucial issues. Therefore, to evaluate read margin (RM) of the array, we calculated RM using the Kirchhoff equation [42]. In detail, when the proposed device is applied to the crossbar arrays, RM depends on read voltage (V_R) of unselected cells called the read scheme [43]–[45]. Among the various read schemes, when the voltage of the unselected bit line (BL) and word line (WL) were 1/2 V_R read scheme, RMs for N_2- and H_2-annealed RRAM cells were evaluated to be 84 and 160, respectively, as shown in Fig. 3(b). Compared to the RM of the N_2-annealed RRAM cell, it is found that the H_2-annealed RRAM cell can more safely protect stored data than the N_2-annealed RRAM cell from sneak current.

To identify the current conduction mechanism of both N_2- and H_2-annealed RRAM cells in detail, the *I-V* curves were re-plotted with several current conduction mechanisms such as Schottky Emission (SE), Poole-Frenkel (P-F) Emission, Space Charge Limited Conduction (SCLC), Fowler-Nordheim tunneling (FNT) and direct tunneling (DT), and trap-assisted tunneling (TAT) [46]–[49].

As a result, it is revealed that the current conduction within Zr_3N_2 films observed in N_2- and H_2-annealed RRAM cells is classified into DT and FNT, according to the magnitude of the applied electric field, as shown in Figs. 4(a) and (b). Especially, enlarged views of FNT fitting regions which colored in green color were indicated in Figs. 4(a) and (b), respectively. First, at both HRS and LRS, when applying a bias to the sample, the initial current generated by DT. Then, when the applied voltage increases, FNT was continuously accompanied due to the band bending, which is consistent with our previous work [27]. Accordingly, this result indicates that the RS of both devices occur by trapping and detrapping of holes within Zr_3N_2 films during the tunneling process. FNT in the H_2-annealed RRAM occurred below -1 V, while in the N_2-annealed RRAM cell, it occurred below -0.56 V. As a result, a hydrogen annealing can enhance the passivation of

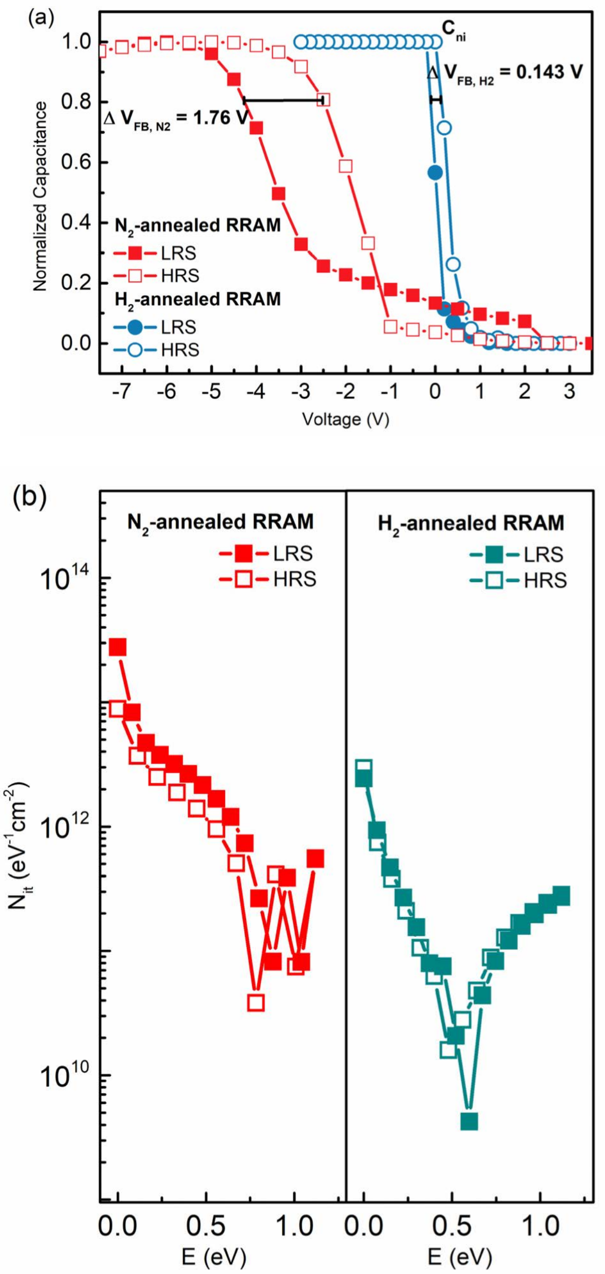

FIGURE 5. (a) *C–V* characteristics of N_2- and H_2-annealed RRAM cells, and (b) N_{it} of LRS and HRS for N_2- and H_2-annealed RRAM cells.

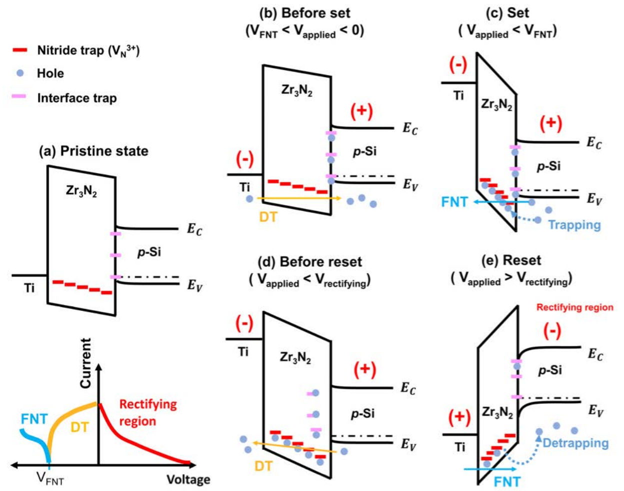

FIGURE 6. Energy band diagram on the conduction mechanism of H_2-annealed RRAM cell. Each schematic indicates (a) pristine state, (b) a state applied voltage below 0 V, (c) set process below V_{FNT}, (d) set process below $V_{rectifying}$, and (e) reset process above $V_{rectifying}$.

interface traps, so a change of barrier height occurred, which ultimately changed the voltage that causes FNT [50]–[52]. As shown in Fig. 3, compared with N_2-annealed RRAM cell, the operation current of H_2-annealed RRAM cell in the *I-V* curve lowered at least 2 order because the passivation of interface traps raise barrier height [53], [54].

Next, to further understand the RS mechanism, we investigated the correlation between traps, N_{it}/N_{nt}, and the RS characteristics of the proposed Zr_3N_2 RRAM. Figure 5(a) shows typical high frequency $C-V$ curve characteristics for the N_2- and H_2-annealed RRAM cells. First, in order to evaluate a change of N_{nt}, we used the Kirchhoff equation, $N_{nt} = -(1/q)C_{ni}V_{FB}$, where C_{ni} is capacitance of nitrides, V_{FB} is shift of flat band voltages. As a result, in the N_2-annealed RRAM, N_{nt} revealed 2.78×10^{12} cm^{-2} at HRS and 4.24×10^{12} cm^{-2} at LRS, and in the case of the H_2-annealed RRAM, N_{nt} were lowered to 1.01×10^{11} cm^{-2} at HRS and 1.38×10^{11} cm^{-2} at LRS, respectively.

Apart from the lowering of the whole N_{nt} level after hydrogen annealing, the difference of N_{nt} in LRS and HRS indicates that RS of the H_2-annealed RRAM cell might be the result of hole trapping and detrapping, which is also consistent with the result of a previous study, such that ΔN_{nt} is due to RS during the set and reset process through charge trapping [27]. In addition, even though the H_2-annealed RRAM cell has smaller ΔN_{nt} than that of N_2-annealed RRAM cell, the CR for H_2-annealed RRAM cell was larger than that for the N_2-annealed RRAM cell. It was because RS for H_2-annealed RRAM cell might be attributed by a small density of charge in an extremely low current level, while the N_2-annealed RRAM cell required a higher charge density to change its resistance due to its high operating current [55], [56].

In addition, to investigate the quantitative analysis of N_{it} in detail, we calculated N_{it} at Zr_3N_2 film/p-Si layer using the Terman method [57]–[62]. As shown in Fig. 5(b), after hydrogen annealing, N_{it} in the whole energy region was lowered, compared to that of the N_2-annealed RRAM cell, which can be explained by hydrogen passivation as shown in the schematic structure of Fig. 1 [63], [64]. As a result, in the Zr_3N_2 RRAM based on the charge trapping/detrapping model, resistance states and the operating voltage was decided by the density of nitride traps, while the decrease of N_{it} results in a reduction of leakage current.

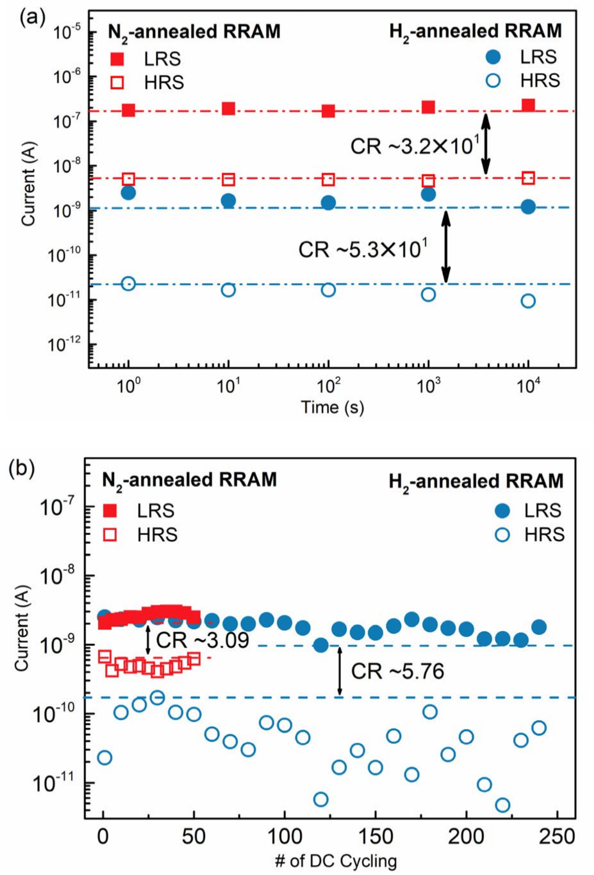

FIGURE 7. (a) Retention characteristics measured up to 10^4 s, and (b) repetitive DC RS characteristics of N_2- and H_2-annealed RRAM cells.

On the basis of the revealed conduction mechanism, such as RS, current conduction, and $C-V$ characteristics, in Fig. 6, we described the conduction process in a band diagram of the Zr_3N_2 RRAM for a more detailed explanation. At the initial state, in Fig. 6(a), current occurs by DT until FNT as shown in Fig. 6(b). Then, when the applied voltage increases, FNT occurs due to band bending at FNT voltage ($V_{FNT} = -1$ V) and holes are trapped in V_N^{3+} as described in Fig. 6(c). As we re-swept a DC bias up to the rectifying voltage ($V_{rectifying}$) of 0.15 V, the LRS state was maintained and DT was the dominant conduction mechanism as shown in Fig. 6(d). Then, when the applied DC bias was the $V_{rectifying}$, the rectifying characteristic was observed. As a result, DT and FNT were dominant conduction mechanisms as same as our previous result [27].

Finally, in order to examine the reliability induced by variation of N_{it} since it degrades a devices' s uniformity with charge trap model in previous results [35], [36], we investigated the retention and endurance characteristics of the N_2- and H_2-annealed RRAM cells. The retention was measured up to 10^4 sec at RT for both RRAM cells, as shown in Fig. 7(a). As a result, LRS and HRS of both cells were stably maintained up to 10^4 s, without any degradation. Then, so as to evaluate endurance for N_2- and H_2-annealed RRAM cells, we investigated repetitive RS characteristics in the dc mode, respectively, as shown in Fig. 7(b). As a result, a CR of 3.09 was maintained for 50 cycles for the N_2-annealed RRAM cell, while the higher CR of 5.76 was observed for 250 cycles for the H_2-annealed RRAM cell. Consequently, hydrogen passivation is a very useful method to lower the N_{it}, which results in the improvement of endurance in charge trap-based RRAM devices.

IV. CONCLUSION

So as to evaluate the H_2-annealing effect on the Zr_3N_2 RRAM devices, we investigated the correlation between traps and RS characteristics. As a result, we observed a decrease of operating voltage and current by lowering N_{it} via the hydrogen passivation process. Even though N_{nt}, which was a dominant factor to decide the resistance states in charge trap-based RRAM devices operating in high current over milli-ampere, was decreased, H_2-annealed RRAM devices exhibited higher CR due to the reduced current level in whole voltage region, resulting in increasing the read margin. In addition, it was found that the reduction of interfacial trap through hydrogen passivation improved the endurance characteristics of the charge trap-based RRAM devices.

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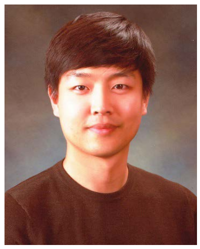
DONGJOO BAE was born in Seoul, South Korea, in 1994. He received the B.S. degree in information and communication engineering from Sejong University, Seoul, where he is currently pursuing the M.S. degree in electrical engineering and convergence engineering for intelligent drone. His research interests include the application of RRAM to CBAs, electrical and electronic engineering, and gas sensors.

DOOWON LEE received the Ph.D. degree in electrical engineering from Sejong University, Seoul, South Korea in 2020. He is currently a Postdoctoral Researcher in electrical engineering with Sejong University. His current research interest includes the electrical/optical semiconductor devices.



SUNGHO KIM received the M.S. and Ph.D. degrees in electrical engineering from the Korea Advanced Institute of Science and Technology (KAIST), South Korea, in 2008 and 2012, respectively. He is currently an Associate Professor of electrical engineering with Sejong University, South Korea. His current research interests include the neuronal and synaptic nano-devices for the neuromorphic systems, at both device and circuit levels.

HEE-DONG KIM received the B.S., M.S., and Ph.D. degrees in electrical engineering from Korea University, Republic of Korea, in 2007, 2009, and 2014, respectively. From 2014 to 2015, he was with the Department Technology, IHP (Leibniz-Institut für innovative Mikroelektronik), Germany, as a Postdoctoral Fellow. He is currently an Associate Professor with the Department of Electrical Engineering, Sejong University, Republic of Korea. His current research interests include electrical/optical semiconductor devices and nonvolatile memory devices.